How much does product classification cost in basis points?

Policy brief: classifying Thai product text for price measurement

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The views expressed are the author's own and do not necessarily reflect those of the Bank of Thailand. This work uses only publicly available data.

Why it matters. Prices from retailers' websites and scanner data can make price measurement faster and more detailed. Before such prices can enter a consumer price index, every product has to be assigned to an item of the official basket: a bag of jasmine rice to rice, a carton of UHT milk to milk and dairy products. Done by hand this does not scale; done by a model it introduces errors. This study asks how large those errors would be in the unit that matters for policy: basis points (hundredths of a percentage point) of measured inflation.

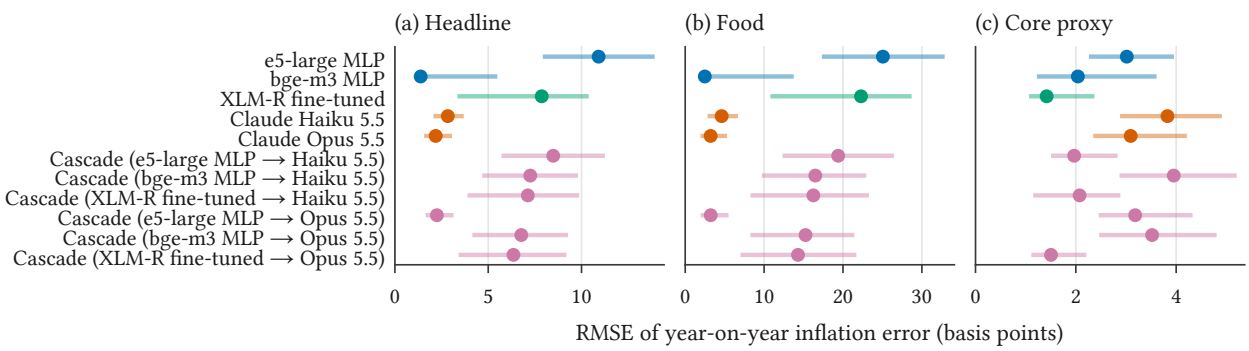
What we did. Using only public data, we tested four kinds of method on Thai product descriptions: low-cost text-matching models, a fine-tuned language model, two commercial large language models (LLMs) instructed with a few examples, and "cascades" that send only the hard cases to an LLM. Each was scored on 3,910 official Thai price-collection descriptions published by the Ministry of Commerce and held out from all training, with every item weighted by its share of household spending. We then fed each method's mistakes through a reconstruction of the Thai CPI to see how far measured inflation would move.

Three findings

1. **The best LLM classifies almost all of the basket correctly; low-cost models do not.** Claude Opus 5.5 assigned 97.9% of the evaluated CPI weight to the right item and Claude Haiku 5.5 95.4%. The best low-cost model reached 69.5%. A cascade that lets a low-cost model handle the easy cases and sends the rest to Claude Opus 5.5 matched Claude Opus 5.5's accuracy at 47% of its cost. All LLM runs for this study together cost USD 14.72.
2. **With the best LLM, classification would move headline inflation by about 2.2 basis points.** That is the typical error of the year-on-year rate over 10 months (at most 2.9 basis points with 95% confidence); for food inflation it is 3.2 basis points. With a low-cost text-matching model the headline error is 10.9 basis points and the food error 25.1; another low-cost model would have understated food inflation in every month.
3. **Accuracy alone is a poor guide, and better documentation beats a better model.** Two models with the same accuracy differed several-fold in basis points, because what matters is whether the confused items' prices moved apart. Most of the LLMs' remaining errors came from two items whose official contents differ from what their names suggest: the official "personal items" include wristwatches, school bags and hair dryers, and margarine is counted with milk and dairy products, not with oils and fats.

What it means for central banks in Asia

- **Ask for basis points, not accuracy.** The method needs only a test set of labelled products, the published CPI weights and the published item indices. Any office can use it to state the classification error of a new data source before the source is used.
- **Publish the specification list and the basket's contents.** The Ministry of Commerce's open specifications made this study possible. A public description of what each item includes and excludes would remove the largest errors found here, for every method.
- **LLMs are affordable at this scale; the issue is governance.** Commercial models change over time. Responses should be stored and versioned, and an office may prefer a model it can run itself.
- **Languages with few labelled data can start from other languages.** With no Thai training labels at all, a low-cost model trained on products in other languages placed 60.3% of the weight correctly. That is a starting point, not a result an index can rely on.



Source: TPSO consumer price indices and weights (base 2566); Open Food Facts (ODbL); TPSO price-collection specifications (official labels; held-out split); author's calculations.

Note: Dots: posterior-mean confusion matrix; lines: 95% interval over 1,000 Dirichlet draws; confusion matrices from the held-out official specifications, products in proportion to CPI weights. Out-of-scope items and items without specifications held at official values. Panels have their own scales.

Figure 1: The LLMs and the cascade to Claude Opus 5.5 would misstate inflation by a few basis points; for headline and food inflation, every low-cost model's interval reaches higher than theirs. Typical error (RMSE) of year-on-year inflation caused by each method's classification mistakes, over 10 months. Dots: central estimate; lines: 95% interval. On core inflation, which excludes fresh food, the low-cost models do as well as the LLMs. Source: Ministry of Commerce; Trade Policy and Strategy Office; author's calculations.

Caveats

The test descriptions are the statistical office's own short texts; shop descriptions are longer and noisier, and accuracy on them will be lower. The basis-point figures assume a mix of products in proportion to household spending, cover the 24.9% of the headline weight that the test set can evaluate, and measure classification error only, not the other errors of a web-price index. A low basis-point error is meaningful only together with high accuracy: a model that classifies little of the basket also moves the index little. The results are for one country and one point in time and are not directly comparable with studies of other markets.

Full paper, code and data: `https://github.com/thanatchon36/cpilens` (DOI 10.5281/zenodo.23686886).